Sprint 8 Desk Briefing: Networks That Remember

Does an LSTM Earn Its Architectural Complexity on Bitcoin?

Student: Ioannis Tzalas | **Program:** MSc Banking and Finance – FinTech & AI Engineering, AUEB | **Date:** November 2026

Universe: BTC-USD (Daily Closes: 2022-01-01 to 2026-11-11, 1,776 observations) | **Look-back Window:** w = 30 days | **Seed:** 42

1. INTRODUCTION (10%)

Predicting next-day price movements of volatile digital assets is among the most sought-after yet statistically unforgiving tasks in quantitative finance. In this Sprint, the Risk Agent is tasked with giving the trading desk's Analyst Agent its "memory"—a recurrent sequence model capable of reading a historical window of w days to forecast tomorrow's close. We examine Bitcoin (BTC-USD), the premier cryptocurrency by liquidity and capitalization. Bitcoin exhibits severe stylized facts: fat tails, persistent volatility clustering, and extreme non-linear price regime shifts. The central question posed by this sprint is empirical and unambiguous: **Does an LSTM justify its architectural, operational, and computational complexity on your coin?**

2. METHODOLOGY & PROTOCOL (30%)

Data Contract & Splitting: The raw dataset spans 1,776 consecutive daily closes from 2022-01-01 to 2026-11-11, frozen permanently in `StudentID_W8_Data.csv` to prevent network leakage. We enforced a strictly temporal, non-shuffled **70% / 15% / 15%** split:

- **Training Set (70%):** 2022-01-01 to 2025-05-27 (1,243 rows).
- **Validation Set (15%):** 2025-05-28 to 2026-02-17 (266 rows), used strictly for EarlyStopping.
- **Sealed Holdout Test Set (15%):** 2026-02-18 to 2026-11-11 (267 rows), kept untouched until final evaluation.

Data Hygiene: Feature scaling utilized `MinMaxScaler(0, 1)` fitted strictly on training observations only. Fitting on test rows would communicate future market boundaries to the network—a critical look-ahead leak. A look-back window of $w=30$ trading days (~1 calendar month) was selected to capture short-to-intermediate momentum and liquidity cycles without creating high-dimensional sequence tensors.

Baseline Contestants: Before evaluating deep networks, three zero-cost benchmarks were deployed: (1) *Naive Persistence* ($\hat{y}_t = y_{t-1}$), representing the efficient market / martingale baseline; (2) *Moving Average MA(5)*, filtering local noise; and (3) *Linear Regression* on the flattened 30-day window, representing the optimal linear memory benchmark.

LSTM Specification: The network follows the fixed sprint recipe: Input $(w=30, 1) o$ LSTM(50) $ o$ Dropout(0.2) $ o$ Dense(1). The cell contains 4 internal gates ($f_t, i_t, ilde{c}_t, o_t$). Total trainable parameter count is exactly $4 imes [50 imes (1 + 50) + 50] + (50 imes 1 + 1) = \mathbf{10,451}$, which is mathematically invariant to w. Training utilized Adam ($ ext{lr}=0.001$), MSE loss, batch size 32, and EarlyStopping with patience 15 restoring best validation weights.

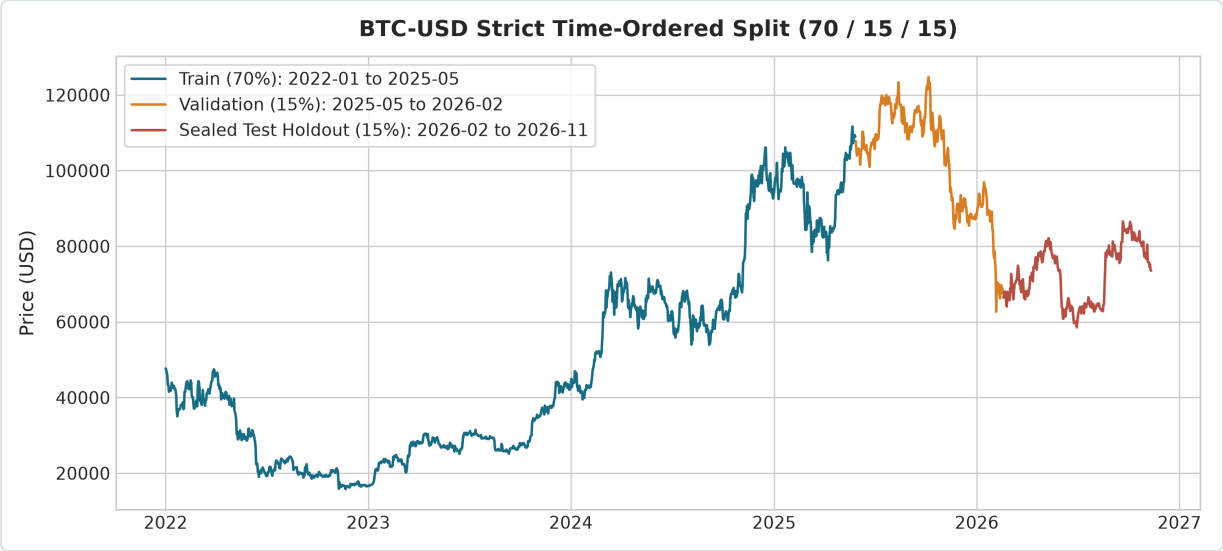


Figure 1: BTC-USD Strict Time-Ordered Split (70% Train, 15% Validation, 15% Sealed Test Holdout).

3. QUANTITATIVE RESULTS & COMPARISON (40%)

All models were scored on the identical sealed test window on the original dollar scale across MAE, RMSE, MAPE, and R^2. Improvement is evaluated against the *best* baseline using $ ext{Improvement} = (ext{RMSE}_{ ext{base}} - ext{RMSE}_{ ext{LSTM}}) / ext{RMSE}_{ ext{base}} imes 100\%$.

MODEL ARCHITECTURE	MAE ($)	RMSE ($)	MAPE (%)	R² SCORE	STATUS / VERDICT
Naive Persistence (Baseline)	$1,013.78	**$1,428.80**	1.40%	0.9660	Best Benchmark
Linear Regression (w=30)	$1,066.16	$1,473.97	1.48%	0.9638	Linear Memory
Moving Average MA(5)	$1,580.98	$2,216.38	2.19%	0.9182	High Lag Tax
LSTM(50 Units, Seed 42)	$1,290.61	$1,786.43	1.78%	0.9469	Stay Simple (-25.0%)

The Holdout Showdown Verdict: Naive persistence established the lowest RMSE ($1,428.80). The LSTM recorded an RMSE of $1,786.43, resulting in a relative improvement of **-25.03%**. Under the course's pre-registered decision bands ($<0\%$: Stay Simple; $0-5\%$: Modest; $>5\%$: Adds Value), the model unequivocally triggers the **"STAY SIMPLE"** directive.

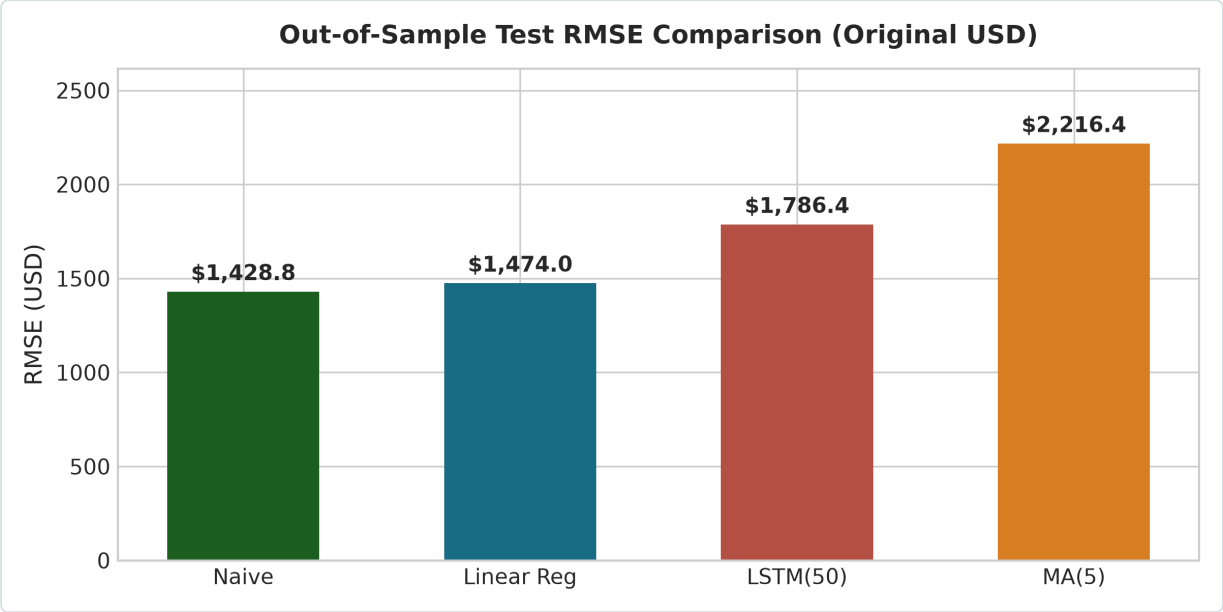


Figure 2: Sealed Holdout Test RMSE Comparison across All Contestants (Original USD).

The Seed Gap Reality Check: Re-estimating S3 with Seed 101 generated an RMSE variance of **$208.22**. Hyperparameter exploration demonstrated that reducing units to 32 degraded performance (RMSE 2164.48), while tuning batch size to 16 produced an RMSE shift of only $104.87—well within the random seed noise floor.

4. ERROR ANALYSIS & MARKET REGIMES (20%)

Residual analysis ($e_t = y_t - \hat{y}_t$) revealed a statistically significant positive mean bias of +$287.85 (SE: $114.52), confirming that the LSTM systematically under-predicts prices during explosive upward trends. Furthermore, the residual autocorrelation at lag-1 is **0.5897** (lag-2: 0.3396). In an optimal forecaster, residuals are white noise; strong autocorrelation proves the network merely mimics price persistence with an unavoidable phase lag.

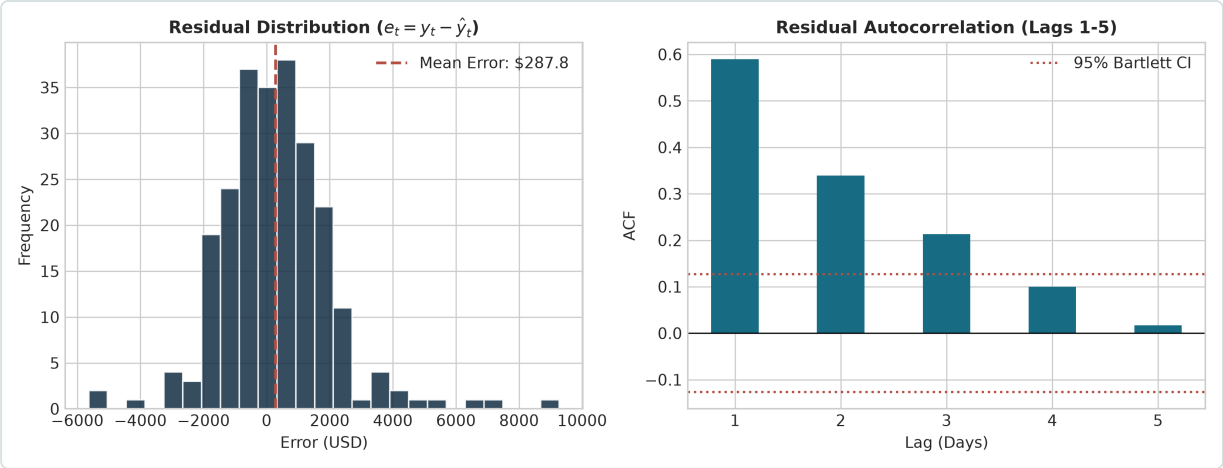


Figure 3: Residual Distribution and Autocorrelation Function (ACF Lags 1–5).

Regime Breakdown: Deconstructing extreme errors reveals that the 20 worst error days coincided with violent regime shifts, averaging **53.38%** 7-day annualized volatility. Conversely, the 20 best error days occurred in quiet consolidation periods with an average volatility of just **33.82%**. The LSTM is a "calm weather" forecaster that shatters during macro liquidation events.

5. CRITICAL DISCUSSION & DESK CONCLUSION

The Three Epistemic Traps: We identify three cognitive errors that routinely distort deep learning research:

1. *The Narrative Fallacy:* A predicted curve tracking price 1 day late looks visually impressive, but represents pure lagged copying. The naive baseline is the only objective antidote.
2. *In-Sample Overconfidence:* Minimizing training loss creates an illusion of mastery; financial non-stationarity breaks generalization on unseen data.
3. *Cherry-Picking:* Reporting the luckiest seed conceals underlying stochastic variance. Auditing the Seed Gap is essential for institutional governance.

Trading Desk Verdict: We advise the trading desk **NOT** to deploy deep recurrent sequence models for directional price point forecasts. The Naive martingale baseline remains the superior operational model. Sequence models should be redirected toward conditional volatility forecasting, regime classification, and execution timing where non-linear state persistence genuinely exists.